SOFTWARE REQUIREMENTS SPECIFICATION

Student Attendance Management System (SAMS)

Version: 1.0

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January 21, 2026

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1 Introduction

1.1 Document Purpose

The purpose of this document is to present a detailed Software Requirements Specifi- cation (SRS) for the Student Attendance Management System (SAMS). This document describes functional and non-functional requirements, system features, constraints, and user characteristics. It provides guidance for developers, testers, administrators, and stakeholders to understand and implement the system.

1.2 Product Scope

The Student Attendance Management System (SAMS) is a web-based platform that al- lows educational institutions to manage student attendance digitally. The system enables teachers to mark attendance online, students to view their records, and administrators to manage users, classes, and subjects. SAMS aims to reduce manual effort, eliminate errors, and improve transparency.

1.3 Definitions, Acronyms, and Abbreviations

- **SAMS:** Student Attendance Management System
- **Admin:** System Administrator
- **UI:** User Interface
- **DBMS:** Database Management System
- **HTTPS:** Hypertext Transfer Protocol Secure
- **OTP:** One-Time Password
- **RBAC:** Role-Based Access Control

1.4 References

Reference	Description
IEEE 830 – Software Requirements Specification Guidelines	This standard provides a basic structure for writing clear and organized SRS documents. It helps ensure that all important system requirements are documented in a proper and understandable format.
ISO/IEC/IEEE 29148 – Requirements Engineering Standard	This standard gives guidelines on how to gather, document, and manage software requirements effectively. It helps maintain consistency and quality in requirement specification documents.
Institutional Attendance Policies and Regulations	These are the official attendance rules set by the educational institution. The system is designed to follow these rules to ensure compliance with academic requirements.

1.5 Document Overview

This Software Requirements Specification (SRS) document is organized into different sections to clearly explain the system requirements.

Introduction: This section explains the purpose and scope of the system. It also includes important definitions, abbreviations, and references to standards used while preparing this document.

Product Overview: This section gives a general idea about the system, its main features, user types, operating environment, and system constraints. It helps readers understand how the system will be used in real situations.

Requirements: This section describes both functional and non-functional require- ments of the system. It explains what the system should do, how it should perform, and the quality of service expected.

Verification: This section explains how the system requirements will be tested and checked. It ensures that the system works as expected and meets all specified require- ments.

Appendices: This section contains additional or supporting information related to the system. It may include references, future considerations, or extra details that help in understanding the document better.

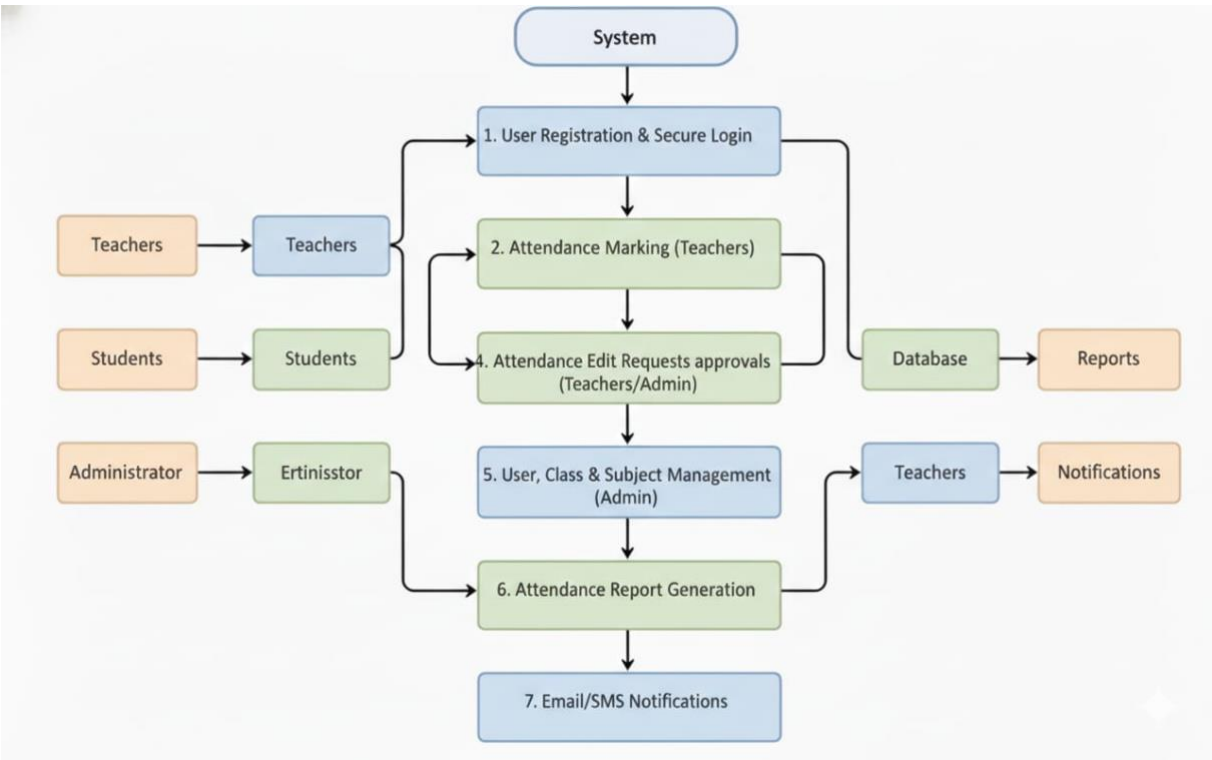
2 Product Overview

2.1 Product Perspective

SAMS is a standalone web-based system that replaces manual attendance registers. It provides centralized attendance management, reporting, and notification capabilities. The system can operate independently but can integrate with institutional databases and email/SMS notification services.

2.2 Product Functions

- **User registration and secure login:** The system allows students, teachers, and administrators to create accounts and log in securely using authentication mecha- nisms to prevent unauthorized access.
- **Attendance marking by teachers:** Teachers can easily mark daily attendance for their assigned classes through an online interface, reducing manual effort and errors.
- **Attendance viewing by students:** Students can view their subject-wise and overall attendance records in real time, ensuring transparency.
- **Attendance edit requests and approvals:** Teachers can request corrections to attendance records, which must be reviewed and approved by the administrator.
- **User, class, and subject management:** Administrators can manage user ac- counts, classes, and subjects to keep the system data organized and up to date.
- **Attendance report generation:** The system can generate attendance reports that can be viewed or downloaded for academic and administrative use.
- **Email/SMS notifications:** The system may send notifications to users regarding attendance updates, low attendance warnings, or important announcements.



2.3 Product Constraints

- **Internet access is required:** The system works only with an active internet connection, as attendance data is stored and accessed online. Without internet access, users will not be able to mark or view attendance.
- **Compliance with institutional attendance rules:** The system must follow all attendance rules defined by the institution, such as minimum attendance percentage and correction policies. Any changes or updates in rules should be reflected in the system accordingly.

- **Secure authentication and role-based access:** Users must log in using valid credentials, and access should be controlled based on their role (student, teacher, or administrator). This helps prevent unauthorized access and ensures data security.
- **Performance constraint:** The system should respond to normal user actions, such as login or attendance viewing, within three seconds. Faster response times are required to provide a smooth and user-friendly experience.

2.4 User Characteristics

- **Students:** Can view attendance records, percentages, and notifications.
- **Teachers:** Can mark attendance, request edits, generate class reports.
- **Administrators:** Can manage users, classes, subjects, approve edits, monitor system usage.

2.5 Assumptions and Dependencies

- Users have basic computer literacy.
- Teachers mark attendance accurately.
- Server uptime is maintained during operational hours.
- Integration with email/SMS gateways for notifications.

2.6 Apportioning of Requirements

- Secure login and role-based access for all user types.
- Management of student, teacher, and administrator profiles.
- Attendance marking, storage, and reporting.
- Notification system for attendance events.
- Periodic reporting and system audit logs.

3 Requirements

3.1 External Interface Requirements

3.1.1 User Interface

The system will provide a web-based UI accessible via modern web browsers. Separate dashboards for students, teachers, and administrators. Forms include input validation and error messages.

3.1.2 Hardware Interface

System accessible via laptops, desktops, or smartphones. Runs on standard server hard- ware.

3.1.3 Software Interface

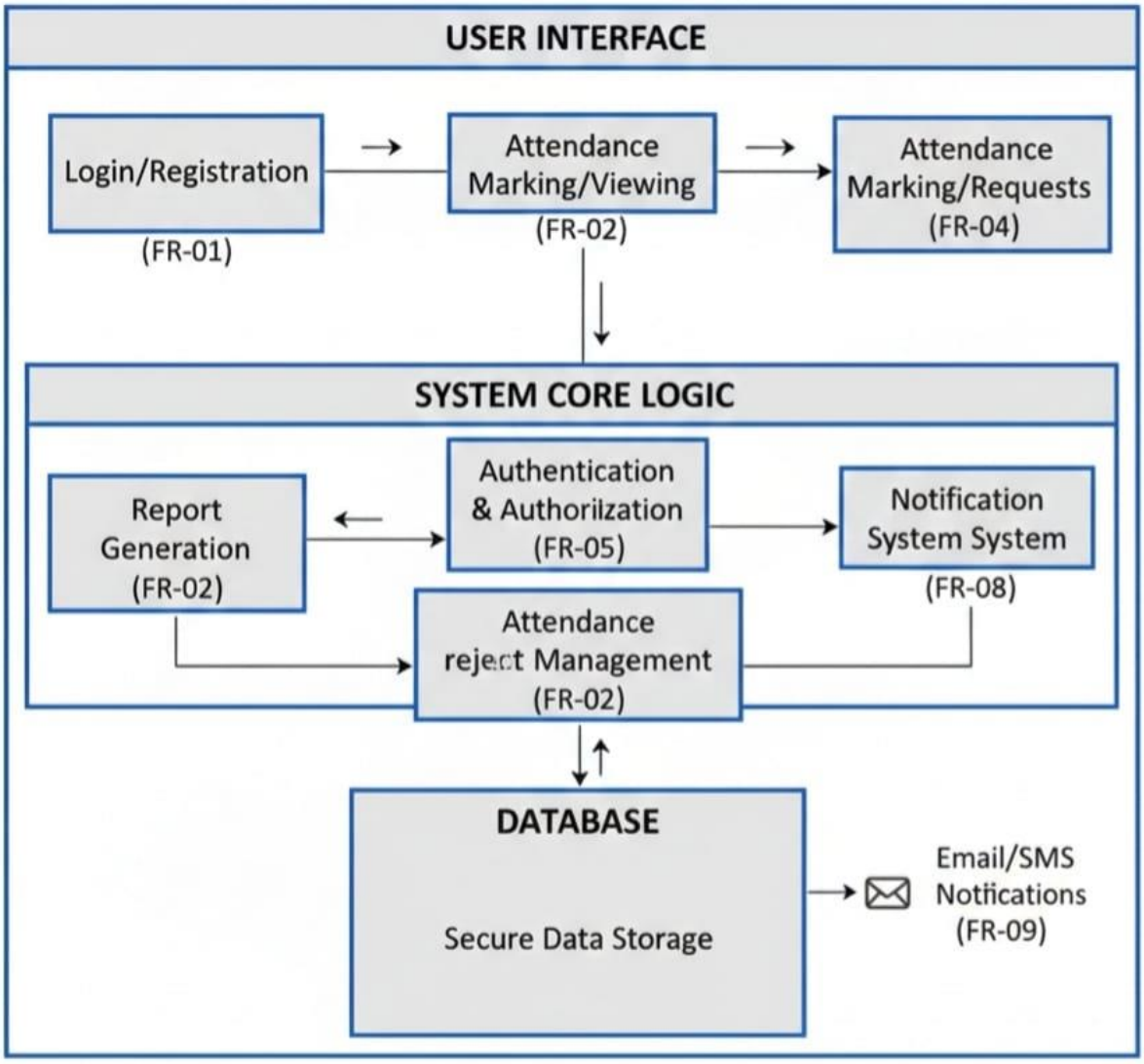
Relational or document-based database for storing user and attendance data. Integration with email/SMS gateways for notifications.

3.1.4 Communication Interface

Secure communication via HTTPS. Internet connectivity is required.

3.2 Functional Requirements

FR No.	Priority	Description
FR-01	High	The system shall allow users to register and log in using valid credentials. Secure login ensures that only authorized students, teachers, and administrators can access the system.
FR-02	High	The system shall allow teachers to mark attendance for their assigned classes on a daily basis. This helps replace manual attendance methods and reduces errors.
FR-03	High	The system shall allow students to view their subject-wise attendance and overall percentage. This helps students stay informed about their attendance status at any time.
FR-04	Medium	The system shall allow teachers to request corrections in attendance if a mistake is identified. This feature ensures flexibility while still maintaining control over attendance data.
FR-05	High	The system shall allow administrators to approve or reject attendance edit requests. This ensures that attendance records remain accurate and secure.
FR-06	High	The system shall store all attendance information safely in the database. Stored data should be protected from unauthorized access or loss.
FR-07	High	The system shall allow administrators to manage users, classes, and subjects. This helps keep the system organized and up to date.
FR-08	Medium	The system shall generate attendance reports for teachers and administrators. These reports can be used for academic review and administrative purposes.
FR-09	Low	The system shall send notifications related to attendance updates or warnings through email or SMS. This feature improves communication but is not critical for basic system operation.



3.3 Quality of Service (Non-Functional Requirements)

3.3.1 Performance

System response time shall be under 3 seconds. Supports at least 500 concurrent users.

3.3.2 Security

Data encrypted and role-based access enforced. Secure password storage and HTTPS communication.

3.3.3 Reliability

Available during institutional hours, with scheduled maintenance. Automatic backup of database.

3.3.4 Usability

Intuitive interface, minimal training required. Clear instructions and feedback provided.

3.3.5 Compliance Requirements

Complies with institutional rules regarding attendance and data privacy. No unauthorized access or commercial misuse of attendance data.

3.3.6 Design and Implementation Constraints

Web-based client-server architecture. Dependency on third-party email/SMS gateways for notifications. Deployment only after administrative verification of user accounts.

4 Verification

System verification will be done through the following methods:

- **Unit Testing:** Verify each module (login, attendance marking, reporting) individ- ually.
- **Integration Testing:** Ensure proper interaction between modules (e.g., atten- dance marking updates student view and report generation).
- **System Testing:** Validate full system functionality according to SRS.
- **User Acceptance Testing (UAT):** Teachers, students, and administrators will validate system usability and functionality.
- **Security Testing:** Verify authentication, role-based access, and data encryption.
- **Performance Testing:** Validate response times and concurrent user support.
- **Regression Testing:** Confirm system remains functional after updates or bug fixes.

5 Appendices

5.1 Glossary

- **Attendance Record:** Stored information of student presence in classes.
- **Dashboard:** Main UI screen providing system options based on role.
- **Notification:** Alert messages sent to users regarding attendance events.
- **Administrator:** User responsible for managing the system.

5.2 Index

- A – Attendance, Administrator
- F – Functional Requirements, Feedback
- N – Notification, Non-functional Requirements
- S – Student, Security, System Testing
- T – Teacher, Testing, Technical Constraints